Lesson Planning

Complete Lesson Plan Book

Class: ICS · Class 11

Subject: Mathematics

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Functions: Domain and Range

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Learning Objectives

• Students will define a function and identify functions from sets of ordered pairs.

• Students will find the domain and range of a function from a table and a formula.

• Students will use the vertical line test to check whether a graph is a function.

Key Concepts

• A function is a rule that gives exactly one output for each input.

• The domain of a function is the set of all possible input values.

• The range of a function is the set of all output values it can produce.

• A set of ordered pairs is a function only if no two pairs have the same first element.

• The vertical line test: a graph is a function if every vertical line meets it at most once.

Teaching Methodology (Step by Step)

1. The teacher introduces a 'function machine': put in a number, the machine doubles it and adds one, and the class works out outputs.

2. The teacher defines a function as a rule with exactly one output for each input.

3. The teacher writes a set of ordered pairs on the board and asks students to check whether any input repeats.

4. The teacher explains domain and range using the function $f(x) = x^2$ with a table of values.

5. The teacher draws three graphs — a line, a parabola and a circle — and applies the vertical line test to each.

6. The teacher gives the class two functions to find the domain and range of, and reviews the answers together.

Classroom Activities

• Domain hunt: students find the domain and range of five functions given as sets of ordered pairs.

• Graph check: students use the vertical line test on four graphs and decide which ones are functions.

• Function machine: one student calls out an input, the 'machine' applies a rule, and the class checks the output.

Assessment Questions

1. Define a function in your own words.

2. Find the domain and range of $f(x) = x^2$ for $x = -2, -1, 0, 1, 2$.

3. Explain why a circle is not the graph of a function.

Homework

• Find the domain and range of $f(x) = 2x + 3$ for $x = 0, 1, 2, 3, 4$.

• Draw the graph of $y = x^2$ and use the vertical line test to show it is a function.

Time Allocation

40 minutes